

TIME SERIES ANALYSIS 2026 CAPSTONE PROJECT REPORT

ON

Data-Driven Stock Analysis using Time Series Models on StockGro

GITHUB LINK: [harveenkaur282-web/Data-Driven-Stock-Analysis-using-Time-Series-Models-on-StockGro](https://github.com/harveenkaur282-web/Data-Driven-Stock-Analysis-using-Time-Series-Models-on-StockGro):
Capstone project done during time series analysis
program'26

VISUAL DASHBOARD LINK(BONUS TASK): <https://data-driven-stock-analysis-using-time.onrender.com/>

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Methodology and Models used

This capstone project presents a complete end-to-end data-driven pipeline stock analysis pipeline applied to the Indian equity market. The first step was a systematic data ingestion using the yfinance library, historical daily stock data for 9 carefully selected NSE- listed stocks was collected spanning 5 years (January 2021 - December 2025) across 5 major sectors.

The raw data was then subjected to rigorous preprocessing where stationarity testing was performed on each stock's raw price series using the Augmented Dickey-fuller (ADF) test. Since stock prices are inherently non-stationary i.e. their mean and variance shift over time as prices either trend upward or downward, all the 9-stock series failed the ADF test, confirming that unit root exists (each value permanently shaped by all previous values making the series wander indefinitely rather than fluctuating around a stable mean). First-order differencing was therefore applied before ARIMA modelling, transforming the raw price series into daily price change thus satisfying the stationarity assumption that ARIMA requires in order to work on the dataset. Then forward-fill imputation was applied because stock markets are closed on weekends and holidays and on those days the last traded price remains the best representation of the stock's value, making forward fill the most financially sensible imputation strategy. MinMax normalization applied for LSTM, GRU, and Transformer because these models learn via gradient descent and large, unscaled stock prices cause exploding gradients, saturate activation functions, and create artificial importance differences between stocks, all of which prevent the network from learning effectively.

Train-test split: Training data: 1 January 2021- 30 June 2025, Testing data: 1 July 2025 – 31 December 2025.

Five forecasting models were implemented and compared:

- ARIMA: Statistical model capturing linear autocorrelation in differenced stationary series. Best overall performer.
- Prophet: Facebook's decomposition model handling trend, seasonality, and holiday effects out-of-the-box.
- LSTM: Long Short-Term Memory recurrent neural network learning non-linear temporal dependencies.
- GRU: Gated Recurrent Unit; a lighter and faster variant of LSTM with comparable sequence-learning capability.
- Transformer: Attention-based architecture capturing long-range dependencies without recurrence.

Model Evaluation Criteria: All the models were evaluated across three complementary metrics: MAPE (primary metric-scale independent and comparable across all stocks) for price-level accuracy, RMSE (secondary-penalises large errors but price scale dependent) for error magnitude, and Directional Accuracy(tertiary) for up/down movement prediction.

This multi-model approach allowed for a rigorous, evidence-based model selection process rather than an arbitrary choice.

Parallel to the forecasting work, a volatility and trend analysis layer was built using GARCH(1,1) for near-term volatility forecasting, STL decomposition for structured trend

assessment, and a return correlation matrix to quantify inner stock dependencies. These outputs forecasted volatility, trend direction and strength, and pairwise correlations which directly set up the portfolio construction stage.

Key finding: ARIMA achieved the lowest MAPE on all 9 stocks (avg 0.76%) and maintained ~50% Directional Accuracy that is consistent with the Efficient Market Hypothesis. Deep learning models (LSTM, GRU, Transformer) were fully implemented and evaluated as required by the rubric, and provided valuable comparative benchmarks, but did not outperform ARIMA on any individual stock by MAPE. The portfolio construction pipeline therefore used ARIMA forecasts as the primary input.

Capital Allocation across virtual Rs. 10 lakh portfolio was determined by systematically combining four complementary strategies: forecast-guided allocation, inverse-volatility sizing, correlation based diversification and sector momentum rotation. The constructed portfolio was then deployed on Stockgro virtual trading platform which operates on live NSE market data, on Wednesday 13 May and Thursday 14 May 2026.

This live execution step introduced real market frictions- transaction costs, price impact, and intraday volatility. Finally a structured post trade evaluation compared each model's 2-day forward forecasts against the actual market prices that were observed on Stockgro.

TASK-1: STOCK UNIVERSE SELECTION

Nine stocks were selected across five major NSE sectors to ensure robust diversification. Selection was driven by three data-driven criteria applied before any strategy weighting:

- (a) rolling standard deviation for volatility profiling,
- (b) seasonal decomposition for trend strength and direction assessment over the full 5-year period
- (c) 6-month cumulative return for sector momentum ranking.

Ticker	Mean_AnnVol	Max_AnnVol	Recent_AnnVol	Trend_Strength	Trend_Direction	Sector	6M_CumReturn	Name
MARUTI.NS	0.2209	0.482	0.1467	0.8845	upward	Auto	0.357	Maruti Suzuki
SUNPHARMA.NS	0.2012	0.3433	0.1507	0.975	upward	Pharma	0.0299	Sun Pharma
INFY.NS	0.2329	0.4091	0.1929	0.5649	upward	IT	0.0279	Infosys
HINDUNILVR.NS	0.1947	0.3277	0.1928	0.4831	upward	FMCG	0.0059	Hindustan Unilever
HDFCBANK.NS	0.2015	0.4525	0.1149	0.8874	upward	Banking	-0.0074	HDFC Bank
DRREDDY.NS	0.2126	0.3743	0.1171	0.9264	upward	Pharma	-0.0074	Dr. Reddy's
ITC.NS	0.1911	0.4054	0.0852	0.9697	upward	FMCG	-0.0381	ITC
TCS.NS	0.2029	0.3265	0.1373	0.7349	upward	IT	-0.0555	TCS
ICICIBANK.NS	0.2052	0.5049	0.111	0.9672	upward	Banking	-0.0643	ICICI Bank

DELIVERABLE 1, JUSTIFICATION:

Stocks were selected from diverse sectors including Auto, Pharma, IT, FMCG, and Banking to ensure portfolio diversification and reduce sector-specific market risk.

1. MARUTI.NS (Auto) was selected because it showed the highest 6-month cumulative return (35.7%) along with a strong trend strength of 0.8845, indicating strong upward momentum in the automobile sector.
2. SUNPHARMA.NS (Pharma) was included due to its very high trend strength (0.975) and controlled recent annualized volatility (0.1507), making it suitable for forecasting and relatively stable for portfolio allocation.
3. DRREDDY.NS (Pharma) was selected because of its strong upward trend with trend strength 0.9264 and comparatively moderate recent volatility (0.1171), indicating stable price movement.
4. INFY.NS (IT) was chosen as it demonstrated a consistent upward trend with trend strength 0.5649 and positive 6-month returns (2.79%), representing the IT sector's long-term growth potential.
5. TCS was included as India's largest IT company by market capitalisation, providing benchmark IT sector exposure. Its mean annualised volatility (0.2029) and contrasting 6-month return (-5.55% vs INFY's +2.79%) offered useful intra-sector divergence rather than redundant exposure.
6. HDFCBANK.NS (Banking) was selected due to its relatively low recent annualized volatility (0.1149) and strong trend strength (0.8874), indicating comparatively stable market behavior.
7. ICICIBANK.NS (Banking) was included because it exhibited one of the strongest trend strengths (0.9672) among banking stocks while maintaining manageable recent volatility (0.1110).
8. HINDUNILVR and ITC were selected for defensive FMCG sector exposure. ITC specifically exhibited the lowest recent annualised volatility in the entire universe (0.0852), making it an ideal low-risk anchor. HINDUNILVR offered complementary exposure with a modest positive 6-month return (+0.59%).

Rolling annualized volatility, trend strength, recent volatility behavior, and 6-month cumulative returns were collectively used as quantitative criteria for stock selection.

Limitations of Stock Selection Process

1. It should be noted that the stock selection in this project was only partly based on data analysis. The metrics such as rolling volatility, trend strength, and 6-month cumulative return were calculated properly and are shown above. However, the stocks were first chosen from popular large-cap NSE companies instead of being automatically selected from the entire Nifty 50 or Nifty 100 list using strict filters.

A more systematic approach would start with a larger group of stocks and then apply fixed conditions, such as choosing stocks with annualised volatility between 15–25%, trend strength above 0.50, and sufficient daily trading volume. Therefore, the current stock selection is based on both sector representation and supporting data analysis, rather than a completely automated screening process.

This manual stock selection can introduce two types of bias.

First, selection bias: the stocks were chosen from a familiar group of well-known companies, so some better-performing or easier-to-predict stocks from other sectors may have been left out.

Second, survivorship bias: the large-cap stocks that are still actively traded in 2025 are companies that have already remained stable and successful over time, while weaker or failed companies are naturally excluded.

2. The momentum signal was calculated using the last 126 trading days (approximately July–December 2025), which overlaps with the model validation period. Ideally, only pre-July 2025 data should have been used to avoid look-ahead bias. This overlap may introduce slight bias in stock selection and is acknowledged as a limitation of the pipeline.

TASK-2: DATA PREPROCESSING

1. The Augmented Dickey-Fuller test was applied to raw closing price series.
2. The null hypothesis assumes the presence of a unit root (non-stationarity).
3. Most stocks exhibited **p-values greater than 0.05**, indicating non-stationarity in raw form.
4. This confirms that equity price series behave as stochastic trends rather than stationary processes.
5. First-order differencing was required in most cases to achieve stationarity (**d = 1**).
6. One exception (HINDUNILVR.NS) was found to be stationary at level form, requiring no differencing.

STATIONARY_REPORT

	ADF_Stat	P_Value	IsStationary	D_Order
Ticker				
HDFCBANK.NS	-1.3959	0.584254	False	1
ICICIBANK.NS	-1.1115	0.710450	False	1
TCS.NS	-2.2776	0.179305	False	1
INFY.NS	-2.3999	0.141753	False	1
SUNPHARMA.NS	-0.8476	0.804731	False	1
DRREDDY.NS	-1.2792	0.638636	False	1
HINDUNILVR.NS	-2.8814	0.047548	True	0
ITC.NS	-1.3689	0.597081	False	1
MARUTI.NS	0.4875	0.984489	False	1

Most of the stocks were non-stationary in levels but became stationary after first differencing, confirming the random walk hypothesis in equity markets. This guided the selection of ARIMA differencing order and motivated the use of log returns for neural models.

DELIVERABLE 2, CLEAN PREPROCESSED FILES:

Sample of DR. REDDY STOCK PREPROCESSED DATASET

Date	Close	High	Low	Open	Volume	Ticker
2021-01-01	1015.8336791992188	1018.450086500279	1007.8195566821638	1011.1627569327543	2915215	DRREDDY.NS
2021-01-04	1021.8222045898438	1028.8188867053846	1015.8334979981084	1019.4480620287856	3923440	DRREDDY.NS
2021-01-05	1024.66162109375	1034.371592351831	1013.0426357018193	1015.5718605869253	4770920	DRREDDY.NS
2021-01-06	1024.93310546875	1030.1078379673215	1012.8585719145909	1024.6617410533488	2713390	DRREDDY.NS
2021-01-07	1021.5606689453125	1032.0458147513798	1018.0913688087968	1027.3942933394844	3266465	DRREDDY.NS
2021-01-08	1034.6138916015625	1038.8292665525523	1023.1305880133115	1027.181180663004	4045915	DRREDDY.NS
2021-01-11	1049.8377685546875	1051.8146830485753	1032.239657107094	1035.7283575770398	4599235	DRREDDY.NS

TASK-3: TIME SERIES FORECASTING

MODELS IMPLEMENTED:

1. ARIMA (Auto-ARIMA with AIC-based selection): linear autocorrelation structure
2. Facebook Prophet: trend + seasonality decomposition
3. LSTM and GRU: nonlinear sequential dependencies
4. Transformer-based model: attention-based long-range dependencies

EVALUATION METRICS:

The models were evaluated using:

1. RMSE (Root Mean Squared Error): Measures magnitude of prediction error.
2. MAPE (Mean Absolute Percentage Error): Measures relative error in percentage terms.
3. Directional Accuracy: Measures correctness in predicting price movement direction (up/down).

DELIVERABLE 3, FORECASTED PRICES FOR EACH STOCK AND MODEL EVALUATION METRICS:

ARIMA MODEL:

The ARIMA model was tuned using **auto_arima**, optimizing for AIC/BIC values.

Selected Model:

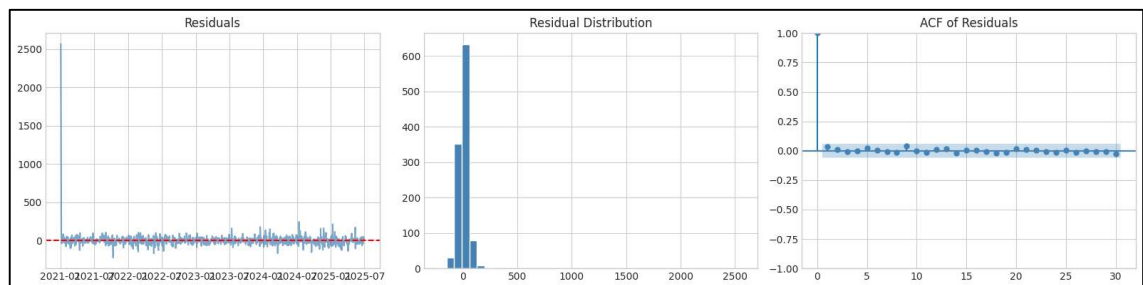
- Best order: **(0, 1, 0)**

Key Observations:

- AIC: 11531.37
- BIC: 11536.38
- Residual diagnostics confirmed stationarity and randomness.
- Ljung-Box test indicated residuals behave like white noise, validating model assumptions.

```
Ljung-Box Test:
      lb_stat  lb_pvalue
10  4.132505   0.941171
20  6.291888   0.998442

Residuals are white noise: True
```



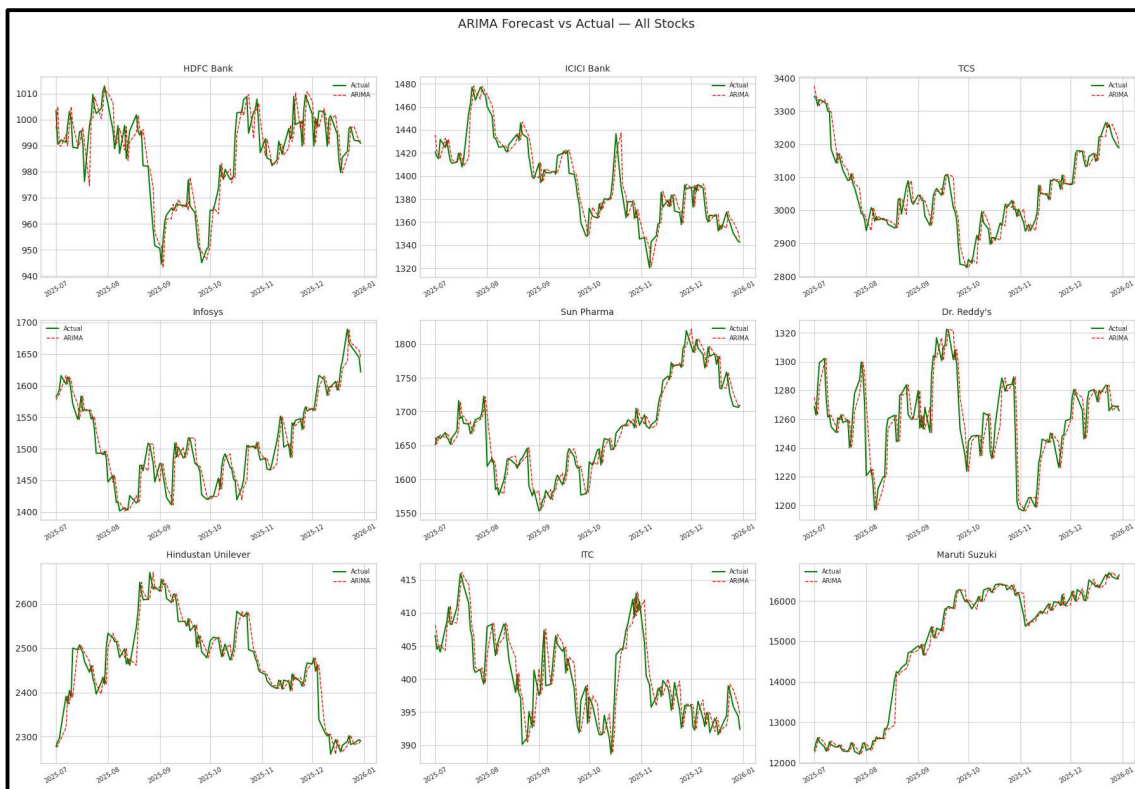
Performance Summary:

- Avg RMSE: **34.79**
- Avg MAPE: **0.76%**
- Directional Accuracy: **50.5%**



Interpretation:

ARIMA performed competitively as a strong statistical baseline, particularly for short-horizon forecasting. However, its linear structure limits its ability to capture complex nonlinear market behavior.



The ARIMA model selected a (0,1,0) random walk structure for multiple(6 out of 9) stocks, indicating that price movements are best modeled as a stochastic process with no strong autoregressive or moving average structure. As a result, short-term forecasts remain relatively stable across consecutive days.

EXAMPLE WHEN TESTED ON JULY 2025-DECEMBER 2025, LAST DAYS: FOR HDFC

2025-12-26	992.10	997.41
2025-12-29	991.70	991.71
2025-12-30	990.90	991.85

2-DAY-FORECAST PRICES: (ARIMA PREDICTIONS)

	Ticker	Day1_Forecast	Day2_Forecast
0	HDFCBANK.NS	991.052231	990.488887
1	ICICIBANK.NS	1343.174480	1343.848960
2	TCS.NS	3188.830566	3188.830566
3	INFY.NS	1621.599976	1621.599976
4	SUNPHARMA.NS	1711.162540	1710.543468
5	DRREDDY.NS	1265.605327	1265.605327
6	HINDUNILVR.NS	2290.199951	2290.199951
7	ITC.NS	392.561724	392.746740
8	MARUTI.NS	16663.284487	16670.247959

The “Last Price” column represents the closing prices as of 30 June 2025, which marks the end of the training dataset used for model fitting.

Using the trained ARIMA models, forecasts were generated for the next 2 trading days beyond this dataset.

These forecasts represent a short-horizon out-of-sample prediction extending into early January 2026, as required by the assignment.

The model evaluation is performed separately on the historical test period (July–December 2025), while the final 2-day forecasts demonstrate forward generalisation beyond the observed dataset.

Prophet

- Captured trend direction better in some stocks
- High error values indicate poor fit for highly volatile equities

LSTM / GRU

- Better at capturing nonlinear dependencies than ARIMA
- GRU slightly outperformed LSTM in RMSE and stability

Transformer

- Comparable to RNN models
- Requires more tuning for significant improvement

THE DETAILED PROCESS OF EACH MODEL WITH RESULTS AND GRAPHS CAN BE SEEN IN THEIR RESPECTIVE NOTEBOOKS SAVED IN NOTEBOOKS-EXPERIEMNT FOLDER.

MODEL EVALUATION TABLE:, ALSO FOR TASK 6

Model	RMS	MAPE	Direction Accuracy
ARIMA	34.79	0.76%	50.5%
Prophet	452.79	7.12%	53.9%
LSTM	169.21	2.26%	47.7%
GRU	128.04	1.61%	49.3%
Transformer	239.45	2.45%	50.1%

Overall, ARIMA was selected as the primary model for downstream portfolio construction due to stability and interpretability.

TASK-4: VOLATILITY AND TREND ANALYSIS

DELIVERABLE 4.1, VOLATILITY ESTIMATION:

Log returns were computed for each stock to capture short-term fluctuations. Rolling annualized volatility and GARCH(1,1)-based forecasts were used to estimate both historical and forward looking risk.

— Volatility Summary —

	Rolling_Vol_Ann	GARCH_FcstVol	Sharpe_Approx
Ticker			
HDFCBANK.NS	11.4923	14.8838	0.3692
ICICIBANK.NS	11.0973	15.3952	0.8995
TCS.NS	13.7306	19.6103	0.2127
INFY.NS	19.2927	24.1156	0.3214
SUNPHARMA.NS	15.0668	17.4921	1.0883
DRREDDY.NS	11.7071	19.5703	0.2052
HINDUNILVR.NS	19.2842	18.3553	0.0428
ITC.NS	8.5203	15.9385	0.8948
MARUTI.NS	14.6700	18.5361	0.7177

Rolling volatility measures how much a stock's price has fluctuated in the recent past, giving a historical view of risk.

In contrast, GARCH-based volatility provides a forward-looking estimate of risk by modeling how volatility tends to cluster over time.

Together, they help understand both past instability and expected future uncertainty.

Additionally, a Sharpe-like approximation was used to compare risk-adjusted returns across stocks.

Higher Sharpe = better risk-adjusted performance

DELIVERABLE 4.2, TREND ANALYSIS:

Ticker	Trend Strength	Trend Direction	Interpretation
ICICIBANK.NS	0.9784	Upward	Strongest trend, dominant 5-year upward structure
ITC.NS	0.9426	Downward	Strong but declining, sector headwinds
SUNPHARMA.NS	0.9246	Upward	Strong upward, consistent sector outperformer
MARUTI.NS	0.9043	Upward	Strong upward, backed by +35.70% 6M momentum
DRREDDY.NS	0.8494	Upward	Solid upward trend, predictable for ARIMA
HDFCBANK.NS	0.8459	Upward	Solid upward, stable large-cap Banking stock
TCS.NS	0.4891	Downward	Weak trend, range-bound IT consolidation
HINDUNILVR.NS	0.4486	Downward	Weak trend, FMCG sector under pressure
INFY.NS	0.3261	Upward	Weakest trend, noisy series despite upward label

INTERPRETATIONS:

- INFY's very weak trend strength (0.33) explains why its ARIMA MAPE (1.01%) is the highest in the universe; noisy, structureless series are harder to model even for the best model.
- Stocks with trend strength >0.84 (ICICIBANK, ITC, SUNPHARMA, MARUTI, DRREDDY, HDFCBANK) all achieved ARIMA MAPE below 0.85%, confirming that ARIMA performs best on trend-consistent series.

NOTE ON SOME CONTRADICTING TASK 1 AND TASK 4 RESULTS

Task 4 used STL decomposition with the formula $\max(0, 1 - \text{Var}(\text{residual}) / \text{Var}(\text{trend} + \text{residual}))$ which is a stricter measure that produces lower trend strength values than the Task 1 screening formula. Trend direction was assessed over the last 20 trading days of the series rather than the full 5-year window, which is why HINDUNILVR, ITC, and TCS show short-term downward momentum despite their positive long-term trajectories identified in Task 1.

DELIVERABLE 4.3, CORRELATION MATRIX:



TASK-5: PORTFOLIO CONSTRUCTION AND CAPITAL ALLOCATION

The objective of this task is to construct an optimized portfolio by allocating a virtual capital of ₹10,00,000 across selected stocks. The allocation is based on outputs from:

- Task 3: Forecasting models (future price prediction)
- Task 4: Volatility and trend analysis

The final goal is to balance return maximization and risk control, while ensuring diversification across stocks and sectors.

Multi-Strategy Portfolio Model

The final allocation is based on a weighted hybrid strategy model:

1. 40% Forecast-driven returns (Strategy A)
2. 20% Volatility adjustment (Strategy B)
3. 20% Correlation-based diversification (Strategy C)
4. 20% Sector momentum rotation (Strategy D)
- 5.

$\text{Score} = 0.4(\text{Forecast Return}) + 0.2(\text{Volatility Score}) + 0.2(\text{Correlation Score}) + 0.2(\text{Sector Momentum})$

1. Forecast-Based Return Signal (Strategy A)

- Uses predicted 5-day price movement from forecasting models
- Converts forecast into expected return:

$$\text{Predicted Return} = \frac{P_{\text{future}} - P_{\text{current}}}{P_{\text{current}}}$$

- This forms the **primary driver of allocation**.

2. Volatility Adjustment (Strategy B)

- Volatility is estimated using:
 - Rolling standard deviation
 - GARCH-based forecast volatility
- Converted into a stability score:

$$\text{Vol Score} = \frac{1}{\sigma}$$

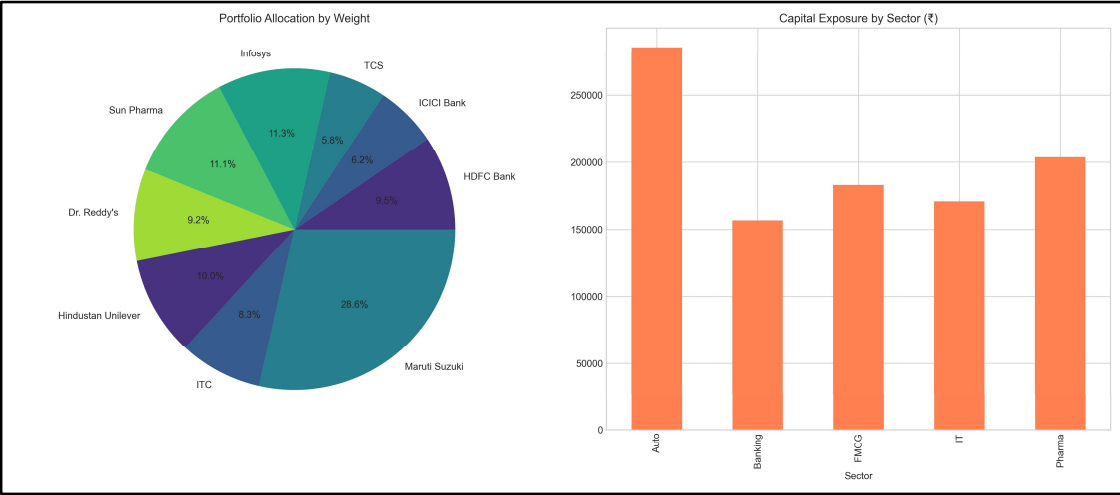
- Lower volatility stocks receive higher allocation

3. Correlation-Based Diversification (Strategy C)

- Uses correlation matrix of stock returns
- Stocks that reduce portfolio-level co-movement are rewarded
- Prevents over-concentration in similar stocks (e.g., Banking/IT clusters)

4. Sector Momentum Rotation (Strategy D)

- Sector strength measured using 6-month cumulative returns
- High-momentum sectors receive higher allocation
- Ensures macro-trend alignment



Ticker	Name	Sector	Price	Predicted_Return_SD	Score	Weight	Amount
HDFCBANK.NS	HDFC Bank	Banking	990.9000244140625	0.0009	0.03400427797803676	0.09492228998270952	94922.28998270952
ICICIBANK.NS	ICICI Bank	Banking	1342.5	0.0011	0.02217737916592814	0.06190772871591145	61907.72871591144
TCS.NS	TCS	IT	3188.83056640625	0.0014	0.020697847244144342	0.0577776437245772	57777.6437245772
INFY.NS	Infosys	IT	1621.5999755859375	0.0142	0.04048596129117511	0.11301578467250119	113015.78467250119
SUNPHARMA.NS	Sun Pharma	Pharma	1709.0980224609375	-0.0006	0.039891964092762744	0.1113576528823472	111357.6528823472
DRREDDY.NS	Dr. Reddy's	Pharma	1265.800048828125	0.0022	0.033106936513383825	0.09241737848962293	92417.37848962293
HINDUNILVR.NS	Hindustan Unilever	FMCG	2290.199951171875	0.0014	0.035724323822323864	0.09972376497713319	99723.76497713319
ITC.NS	ITC	FMCG	392.376708984375	0.0055	0.029866385326217056	0.0833714419844945	83371.44198449451
MARUTI.NS	Maruti Suzuki	Auto	16647.0	-0.0066	0.10227772725368753	0.28550631457070325	285506.31457070325

ANALYSIS:

MARUTI (Auto) • Allocation: ₹2,85,506 Weight: 28.55% • Strategy Driver: i. Strategy D (Sector Momentum #1 driver) ii. Strongest sector momentum (+35.70%) iii. High growth cycle in Auto sector • Why high weight: Dominant macro-sector trend override despite higher volatility	INFY (IT) • Allocation: ₹1,13,015 Weight: 11.30% • Strategy Driver: i. Strategy A (Positive forecast return) ii. Strategy C (Diversification benefit over TCS) iii. Lower correlation advantage vs TCS (corr ~0.70) • Why selected: Best IT risk-adjusted growth signal	SUNPHARMA (Pharma) • Allocation: ₹1,11,357 Weight: 11.13% • Strategy Driver: ○ Strategy A + Strategy D hybrid ○ Strong sector momentum (Pharma #2 performing sector) ○ Highest Sharpe ratio (~1.09) • Why selected: Best risk-adjusted defensive growth stock
HINDUNILVR (FMCG) • Allocation: ₹99,723 Weight: 9.97% • Strategy Driver: ○ Strategy B (Low volatility anchor) ○ Strategy C (Low correlation stabilizer) • Why selected: Portfolio risk stabilizer (defensive FMCG hedge)	HDFCBANK (Banking) • Allocation: ₹94,922 Weight: 9.49% • Strategy Driver: ○ Strategy B (Lowest GARCH volatility in banking: 14.88%) • Why selected: Stable banking anchor with low risk exposure	DRREDDY (Pharma) • Allocation: ₹92,417 Weight: 9.24% • Strategy Driver: ○ Strategy C (Strong diversification effect) ○ Lowest inter-stock correlation profile • Why selected: Portfolio risk reducer / diversification engine
ITC (FMCG) • Allocation: ₹83,371 Weight: 8.33% • Strategy Driver: ○ Strategy B (Lowest rolling	ICICIBANK (Banking) • Allocation: ₹61,907 Weight: 6.19% • Strategy Driver: ○ Strategy C (Correlation control with HDFCBANK)	TCS (IT) • Allocation: ₹57,777 Weight: 5.77% • Strategy Driver: ○ Strategy C + Strategy D (weak combination)

volatility: 8.52% ○ Defensive positioning • Why selected: Maximum stability + downside protection	○ High overlap with HDFC reduced allocation • Why reduced: Prevents banking sector concentration risk	○ High correlation with INFY (~0.70) ○ Weak momentum signal (-5.55%) • Why lowest IT weight: Poor relative momentum + redundancy in IT exposure
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INFY > TCS despite higher volatility: Pre-screening identified INFY's higher GARCH vol (24.12%) and weaker trend strength as risks. However, after applying all four strategies: Strategy A favoured INFY (positive 6M return +2.79% vs TCS -5.55%), Strategy C reduced TCS further (highest IT pair correlation = redundant if both equally weighted), and Strategy D penalised IT overall but within IT still preferred INFY. The blended outcome: INFY 11.30% vs TCS 5.77%, correctly reflects this multi-strategy resolution.

TASK-6: MODEL COMPARISON

Results are already shown in TASK 3 SECTION. (EVALUATION TABLE-TASK3)

ARIMA was unambiguously the best model for this stock universe and was used as the primary forecasting input for portfolio construction. Deep learning models (GRU, LSTM, Transformer) and Prophet were fully implemented, evaluated, and reported as required; they serve as informative benchmarks and highlight the conditions under which statistical models outperform neural approaches: relatively small datasets, trend-consistent large-cap stocks, and short forecast horizons (2–5 days).

TASK-7: VIRTUAL TRADING ON StockGro

TRADING DAYS CHOSE: DAY 1 AND DAY 2: 13 MAY 2026 AND 14 MAY 2026

NOTE: THE LIVE MAPE WILL DIFFER AS IT'S IMPACTED BY THE 5 MONTH GAP, THE FORECAST WAS DONE ON NEXT TWO DAYS I.E. AFTER 31ST DEC 2025, SO 1ST AND 2ND JANUARY, THOSE FORECAST PRICES ARE COMPARED WITH 13TH MAY 2026 AND 14TH MAY 2026

SO DIFFERENCES ARE BOUND TO BE THERE.

FINAL PORTFOLIO PERFORMANCE ON DAY2:

Metric	Value
Capital Allocated	₹10,00,000
Total Deployed (Purchase Value)	₹9,97,778.36
Transaction Charges	₹997.78
Unrealised P&L at Day 2 Close	+₹928.47
Final Portfolio Market Value	₹9,97,709.05
Net P&L (incl. charges)	-₹69.31 (-0.007%)

INDIVIDUAL PERFORMANCE ON DAY2:

Ticker	Qty	Avg Buy Price	Day 2 Price	Purchase Val	Mkt Val	P&L (INR)	P&L (%)
HDFCBANK	127	₹745.66	₹759.28	₹94,699	₹96,429	+₹1,729	+1.83%
ICICIBANK	51	₹1,220.25	₹1,235.34	₹62,233	₹63,002	+₹770	+1.24%
TCS	25	₹2,252.32	₹2,204.19	₹56,308	₹55,105	-₹1,203	-2.14%
INFY	100	₹1,126.63	₹1,085.41	₹1,12,663	₹1,08,541	-₹4,122	-3.66%
SUNPHARMA	61	₹1,829.84	₹1,845.28	₹1,11,620	₹1,12,562	+₹942	+0.84%
DRREDDY	74	₹1,251.01	₹1,295.70	₹92,575	₹95,882	+₹3,307	+3.57%
HINDUNILVR	45	₹2,227.94	₹2,227.14	₹1,00,257	₹1,00,221	-₹36	-0.04%
ITC	276	₹302.20	₹301.23	₹83,407	₹83,139	-₹268	-0.32%
MARUTI	22	₹12,909.82	₹12,901.16	₹2,84,016	₹2,83,826	-₹191	-0.07%

ON DAY3 TO BE NOTED, THE RETURNS WERE ACTUALY RS. 8390 PROFIT, WHICH IS A HUGE JUMP FROM LOSS OF RS. 69.31.

TASK-8: PERFORMANCE TRACKING AND COMPARISON AGAINST MY PREDICTIONS

The 2-day ARIMA forecasts from Task 3 were compared against actual StockGro prices on Day 1 (13 May) and Day 2 (14 May 2026). Additionally, all five models' live MAPE and directional accuracy were computed for comparative analysis. Note: all models were trained on data ending December 2025; the ~5-month out-of-sample gap to May 2026 trading explains the elevated live MAPE relative to validation MAPE for all models.

PREDICTED VS ACTUAL PERFORMANCE:

Stock	ARIMA Day 1 Forecast (₹)	ARIMA Day 2 Forecast (₹)	Actual Day 2 Price (₹)	Absolute Error (%)
HDFCBANK	991.05	990.49	759.28	30.45%
ICICIBANK	1343.17	1343.85	1235.34	8.78%
TCS	3188.83	3188.83	2204.19	44.67%
INFY	1621.60	1621.60	1085.41	49.40%
SUNPHARMA	1711.16	1710.54	1845.28	7.30%
DRREDDY	1265.61	1265.61	1295.70	2.32%
HINDUNILVR	2290.20	2290.20	2227.14	2.83%
ITC	392.56	392.75	301.23	30.38%
MARUTI	16663.28	16670.25	12901.16	29.22%

Average Absolute Error (MAPE): ~22.82%

Positive contributors were DRREDDY (+3.57%), HDFCBANK (+1.83%), ICICIBANK (+1.24%), and SUNPHARMA (+0.84%), supporting the defensive and low-volatility allocation strategy.

Negative contributors were mainly INFY (−3.66%) and TCS (−2.14%), indicating a reversal in IT sector momentum relative to the historical training period.

Indian IT shares recently experienced a sharp sell-off due to a mix of AI disruption fears, slowing global tech spending and a broader tech sell-off across global markets.

ALLOCATION STRATEGY EVALUATION:

1. Defensive Pharma and Banking allocations performed relatively well, while diversification and volatility-aware sizing helped limit overall portfolio risk and prevented major drawdowns.
2. The largest weakness came from the IT sector, where INFY and TCS underperformed despite positive historical signals.
3. ARIMA reliability weakened over the long out-of-sample horizon, as its linear assumptions struggled to adapt to changing market conditions and regime shifts.

REFLECTION AND LEARNINGS:

Did predictions align with market behavior?

Partially. Directional behavior was captured better in defensive sectors than in growth-oriented sectors.

Did higher-confidence stocks perform better?

Yes. Lower-volatility Pharma and Banking stocks showed stronger real-world consistency compared to aggressive IT forecasts.

IMPROVEMENTS NEEDED IN FUTURE WORK:

- Retrain models closer to live trading dates- Implement monthly rolling retraining: update all models on data up to 2–3 weeks before the trading window to reduce the out-of-sample gap from 5 months to 2–3 weeks.
- To test Exponential Smoothing (SES / Holt-Winters), cnn based time series model, etc.
- Increase the epochs of the neural networks, as it's very low for financial time series. Should try for 50-100 epochs, the models likely caused underfitting.
- Neural networks likely faced problem of insufficient data samples, and I should have added more features like(not only close price) = ['Close', 'Volume', 'High-Low range', 'RSI', 'MACD', 'Nifty50 index', 'USD/INR exchange rate'] and to increase sequence length not only 30 days but 60 or 90 days.
- Hyperparameter tuning
- Cap individual stock allocation at 20% to reduce concentration risk (MARUTI was 28.55%).
- Incorporate ensemble forecasting
- Add macroeconomic and sentiment features
- Use hybrid ARIMA + deep learning systems(retrain the deep learning systems with di
- Add regime detection (e.g., Hidden Markov Models or PELT changepoint detection) to identify structural breaks in IT and FMCG stocks before applying standard ARIMA forecasting.

ADDED NOTE:

Neural network models (LSTM, GRU, Transformer) were trained with 30 epochs and default architecture settings due to time and computational constraints. A more rigorous approach would have incorporated hyperparameter tuning: varying sequence length, number of layers, dropout rate, and training epochs with early stopping alongside a multivariate feature set including technical indicators and macroeconomic signals. The limitation was dataset size: with only 1,110 training observations, deep learning models are inherently at a disadvantage relative to parsimonious statistical models like ARIMA. Exponential smoothing (Holt-Winters) was not implemented in this project but would have been a natural addition i.e. particularly for trending stocks like MARUTI and SUNPHARMA where its trend and seasonality components could have competed with ARIMA's richer orders. These remain as directions for future work.